XSuM: a generative surrogate of the SBND reconstruction chain trained on the BNB Monte Carlo — technical note

L. Wan*

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Abstract

XSuM is a per-event surrogate of the SBND simulation and reconstruction chain: given the true neutrino vertex and the post-FSI final-state particles of an interaction, it samples the record that the cafpyana analysis dataframes hold for that interaction (whether and how many Pandora slices are truth-matched to it, the slice-level block of about fifty reconstructed quantities, and later the set of particle-flow objects), in the same dataframe format as the production. Analyses then run unchanged on the surrogated dataframes. The model is trained on the 2026 gen2 BNB + cosmics central-value Monte Carlo (v10_14_02_03) processed with a selection-free cafpyana configuration written for this purpose. This note documents the data, the inputs and outputs, the baseline and flow-matching models, and their closure on held-out files. It is updated as the production and the models progress.

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*lwan@fnal.gov

1 Purpose and scope

The physics case is forward folding without re-simulation: a cross-section model that was never run through Geant4, Pandora and cafpyana (another generator, a modified FSI model, a new kinematic region) is transported to reconstruction level by the surrogate and compared with data or with the nominal MC at that level. The surrogate does not fix data/MC disagreement; it reproduces the MC detector response, including its flaws, for new inputs.

Scope decisions (2026-10-16): XSuM generates the complete reconstruction record, not analysis quantities. Selection flags, BDT scores and derived kinematics such as $|t|$ are recomputed downstream by the analysers (CC coherent pion: S. Oh; charged kaons: T. Wester) from the generated variables. Cosmic-only slices do not depend on the neutrino model and are to be taken from the MC; they are not yet in the surrogated files. Generator-internal labels (interaction mode, CC/NC, W, Q^2, E_ν, signal flags) are never inputs; they are kept in a separate label table for holdout tests.

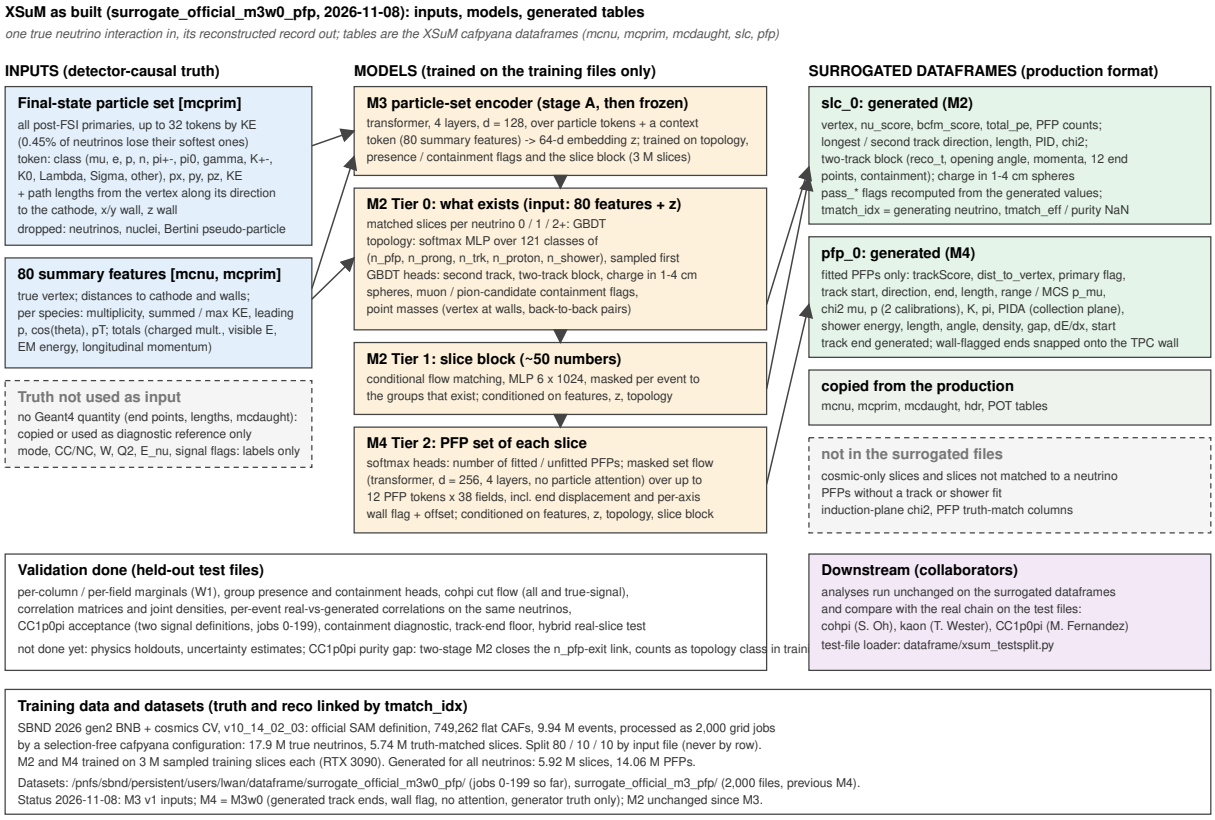


Figure 1: XSuM as built: inputs, the chain of models (M3 encoder, M2 heads and flow, M4 set flow), what the surrogated files contain (generated, copied, missing), the validation done so far and the downstream use. Only implemented components are drawn; the remaining plans are listed in Sec. 10.

2 Data

Production. A cafpyana df-maker configuration (`analysis_village/xsum/`) writes, per flat CAF: `mcnu` (every true neutrino with the full `make_mcdf` truth, the cohpi kinematics and cate-

gory), `mcprim` (every post-FSI primary), `mcdaught` (the Geant4 daughters of every primary), `slc` (every non-clear-cosmic slice with the cohpi variables and each selection stage as a flag), `pfp` (every PFP of those slices with track, shower and PID variables, kaon PID on all planes, parent links and truth match), `hdr` and POT tables. Two productions:

- **scratch1633**: the 1,633 merged flat CAFs (2.5 GB, about 3,300 triggers each) still present on the dCache scratch area referenced by the cafpyana sample list (1,363 of the 2,996 had been purged), run in pool mode on `sbndbuild03` with 30 workers in about 3.5 h: 15.5 GB of dataframes, 6,000,000 neutrinos in the five chunks used here. This production predates the `mcdaught` table, the kaon PID branches and the removal of the 50 cm vertex-distance cut on PFPs.
- **official**: the SAM definition `aurora_SBND2026A_gen2_BNBLight_prodgenie_corsika_proton_rockbox0p1_sl` (749,262 flat CAFs, 9.94 M triggers, 7.9 TB), submitted as 2,000 grid jobs on 2026-10-16 after a 375-file pilot (21.5 s per file single-threaded, 4,500 CPU-hours in total).

A defect of the official production and its repair. The module version used on the grid counted showers among all PFPs of a slice instead of those within 50 cm of the vertex, so the two-track topology never fired and the slice two-track block and the selection flags are empty in the archive. All of them are functions of the PFP table, so they were rebuilt offline exactly as cafpyana computes them (`xsum.prep.repair`) and stored as sidecar `slc` files; the training tables and the surrogated dataset use the repaired tables. The module is fixed for future productions.

Truth–reco link. A slice points to its truth-matched neutrino through `tmatch_idx`; a PFP points to the true particle it was matched to and, through `interaction_id`, to its neutrino. A neutrino can have zero, one or several matched slices; the table keeps the slice with the highest match efficiency and records the count.

Splits. Train / validation / test are 80 / 10 / 10 by input file, never by row. In the five scratch chunks: 4,800,348 / 600,149 neutrinos in train / test. Holdout tests use physics regions (momentum transfer, energy, angle, multiplicity, TPC), see Sec. 10.

3 Inputs and targets

Inputs. The post-FSI primary particles (species, (p_x, p_y, p_z); neutrinos and nuclear remnants dropped) and the true vertex. The M1 and M2 models use 80 fixed features derived from them: the vertex and its distances to the cathode and the walls; per species (mu, e, p, n, $\pi^{\pm}$, π^0, gamma, $K^{\pm}$, K^0, Lambda, Sigma) the multiplicity above the cohpi thresholds (27 / 30 / 20 MeV for mu / pi / p), the summed and maximum kinetic energy and the leading particle's momentum, $\cos\theta$ and p_T; and totals (charged multiplicity, visible and total kinetic energy, electromagnetic energy, longitudinal momentum). These summary features carry no azimuth; the particle-set encoder M3 (Sec. 7) adds a learned embedding of the full list of final-state particles, momentum vectors included.

Targets. Tier 0: the number of matched slices and, per slice, which groups of variables exist (a second track, the two-track topology, hits within 1–4 cm of the vertex) and the two containment flags. Tier 1: the slice block (Table 5 lists the columns). Tier 2 (M4): the PFP set.

4 M1 baselines

Gradient-boosted classifiers for the discrete targets and mixture density networks (MDN, eight diagonal Gaussians, against a single Gaussian) for continuous responses, on the 80 features. Results on the test files of the official sample (reports/m1_official); the scratch-sample results are in the git history of this note.

Slice efficiency. 32.1% of all true neutrinos (dirt and out-of-volume included) have a truth-matched slice. Log loss 0.1822 against a marginal 0.6282, AUC 0.977. The calibration against vertex z, muon momentum, visible energy and multiplicity is shown in Fig. 2.

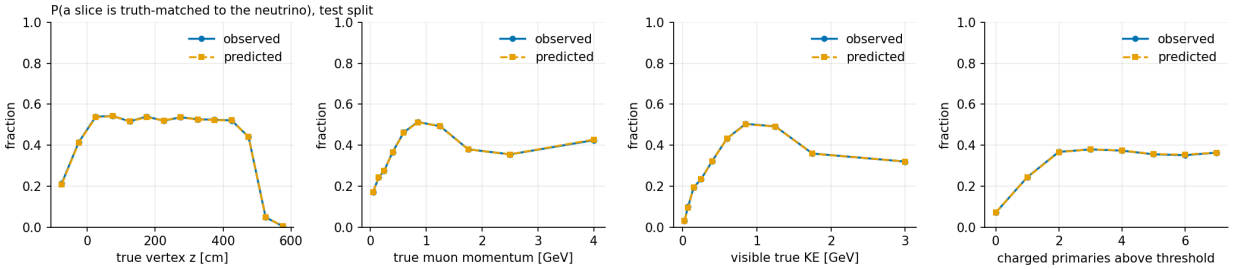


Figure 2: Observed and predicted probability that a slice is truth-matched to the neutrino, test files.

cohpi selection as a yardstick. The full cohpi selection (including the sphere-charge and cathode-crossing cuts) keeps 0.034% of neutrinos. As a check of the production, the selected count per POT agrees with the technote cut flow within 10%. The classifier reaches log loss 0.00275 against 0.00273 (AUC 0.780); per true category:

True category	test neutrinos	selected [%]	predicted [%]
out of FV	492,367	0.000	0.025
NC	30,344	0.053	0.032
signal	149	22.148	37.118
CCQE	40,886	0.024	0.022
2p2h	10,703	0.056	0.026
RES	21,548	0.427	0.321
DIS	4,152	0.554	1.026

PFP multiplicity of the matched slice: log loss 1.087 against 1.561, accuracy 55.7% against 31.8% (Fig. 3).

Continuous responses. Table 1 gives the test negative log-likelihood and Table 2 the marginals; Figs. 4–6 show them.

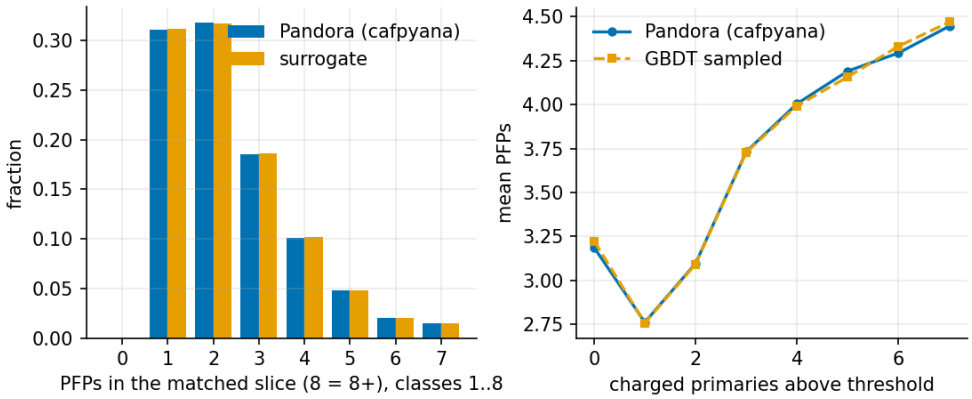


Figure 3: PFP multiplicity of the matched slice: marginal and mean versus charged primaries.

Target	training rows	Gaussian NLL	MDN ($K = 8$) NLL
vertex residual (3-d)	727,641	7.631	1.264
leading track: $\Delta \cos\theta$, log length	709,214	0.054	-2.501
two-track: log range-p ratios, $\Delta \cos\theta_\mu$	13,435	0.744	-1.608

Table 1: M1 MDN test NLL per target.

Variable	Pandora median [16%, 84%]	MDN median [16%, 84%]	W_1
dvtx_x [cm]	0.001 [-0.202, 0.213]	-0.001 [-0.205, 0.210]	2.3311
dvtx_y [cm]	0.002 [-0.378, 0.400]	0.001 [-0.364, 0.388]	3.0223
dvtx_z [cm]	-0.012 [-0.397, 0.488]	-0.010 [-0.376, 0.463]	3.8706
long_dirz - lead cos theta	-0.001 [-0.048, 0.061]	-0.002 [-0.050, 0.059]	0.0636
log(long track length [cm])	4.730 [3.674, 5.494]	4.728 [3.655, 5.501]	0.0086
log(range_p_mu / p_mu_true)	-0.015 [-0.690, 0.082]	-0.016 [-0.636, 0.148]	0.0355
log(range_p_cpi / p_pi_true)	-0.488 [-1.098, -0.132]	-0.514 [-1.074, -0.112]	0.0428
long_dirz - cos theta_mu	-0.000 [-0.030, 0.059]	0.000 [-0.033, 0.060]	0.0303

Table 2: Marginals of the MDN samples against Pandora, test files.

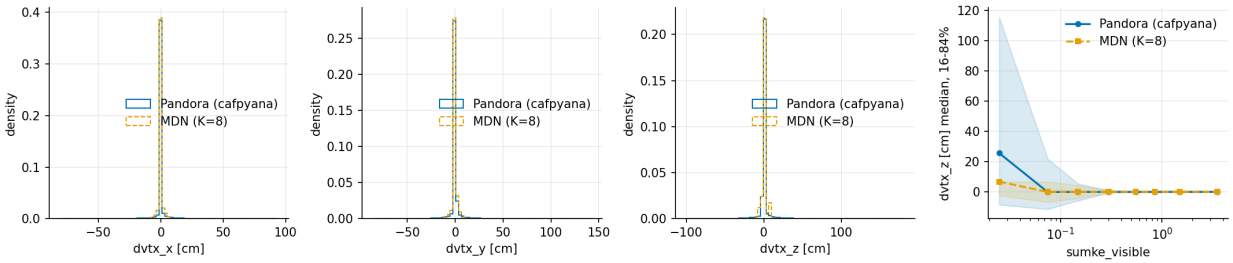


Figure 4: Vertex residual of the matched slice (FV neutrinos).

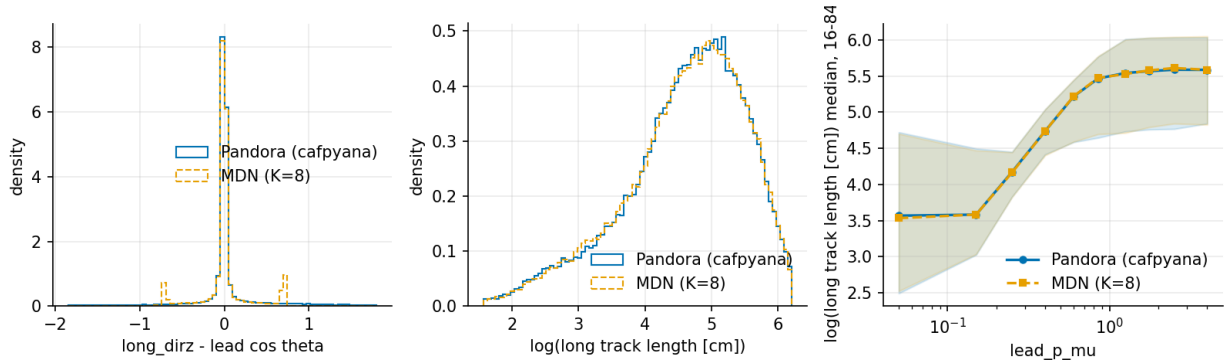


Figure 5: Longest track: direction relative to the leading charged primary and length.

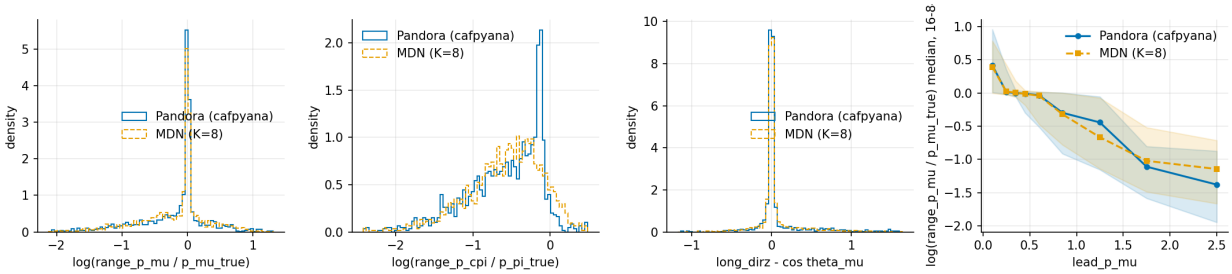


Figure 6: cohpi two-track response on 1mu1pi events passing the two-track topology.

5 M2: flow-matching surrogate of the slice block

Generation proceeds in three steps. (i) A *topology* head: a softmax classifier over the joint pattern of PFP counts (`n_pfp, n_prong, n_trk, n_proton, n_shower`; the 120 most frequent tuples plus "other", whose members are drawn from the training pool), sampled first; its one-hot encoding conditions everything that follows. (ii) Gradient-boosted heads for the presence of the variable groups that may not exist in a slice (second track, two-track block, hits within 1–4 cm of the vertex), for the two containment flags, and for the point masses at detector boundaries (vertex pinned at $z = 0$ or at the x, y walls, back-to-back track pairs). (iii) A conditional flow-matching model (linear path, Gaussian source, 6-layer velocity MLP of width 1024) over the standardised continuous block, masked per event to the groups and dimensions that exist (sim2reco `dim_mask`). Positive quantities enter as logarithms, direction cosines and the opening angle through artanh (and direction vectors are renormalised), standardisation is robust (median, MAD). Inverse transforms produce a dataframe with the production column names. Trained on 5,736,854 matched slices of the official sample for 60 epochs (54 min on one RTX 3090). Host memory is kept near 10 GB by splitting the table once on disk into a matched-only table and a 4 M-row subsample for the cardinality head, training the presence heads on 2 M-row subsamples and building the topology one-hot per batch on the GPU.

Earlier versions and what they taught: v1 (unbounded cosines, counts as dequantised flow dimensions) closed the cohpi cut flow only to 45% in the signal region; v2 (bounded cosines, point-mass heads) to 74%, losing most at the exact-count stage; v3 (topology head) closes every stage within statistics on the scratch sample and to better than 10% at every stage on the official sample (Table 4).

Group / flag	rate [%]	log loss (marginal)	log loss (GBDT)	AUC
short	60.9	0.6693	0.0858	0.995
two	5.2	0.2053	0.0172	0.999
s1	86.8	0.3909	0.3298	0.776
s2	95.3	0.1907	0.1491	0.832
s3	97.2	0.1275	0.0938	0.864
s4	98.1	0.0949	0.0665	0.885
is_muon_contained	43.3	0.6842	0.4050	0.892
is_cpi_contained	80.9	0.4878	0.2879	0.908
spike_vtx_z	13.0	0.3855	0.0948	0.985
spike_vtx_x	5.5	0.2131	0.0507	0.989
spike_vtx_y	5.5	0.2117	0.0623	0.982
spike_opening_angle	2.6	0.1189	0.0875	0.872

Table 3: Group presence and containment heads, test files.

Closure on the test files. Table 5 compares every generated column with Pandora (fraction present and Wasserstein-1 in transformed units; median W_1 0.014, maximum 4.727). Table 4 recomputes the cohpi cut flow on the generated and the real test slices, for all slices and for true signal; this is the kind of comparison the analysers will make on the full surrogated sample.

Correlations. A surrogate that matched every marginal but sampled the variables independently would fail any analysis that cuts on several of them. Fig. 8 compares the Spearman correlation

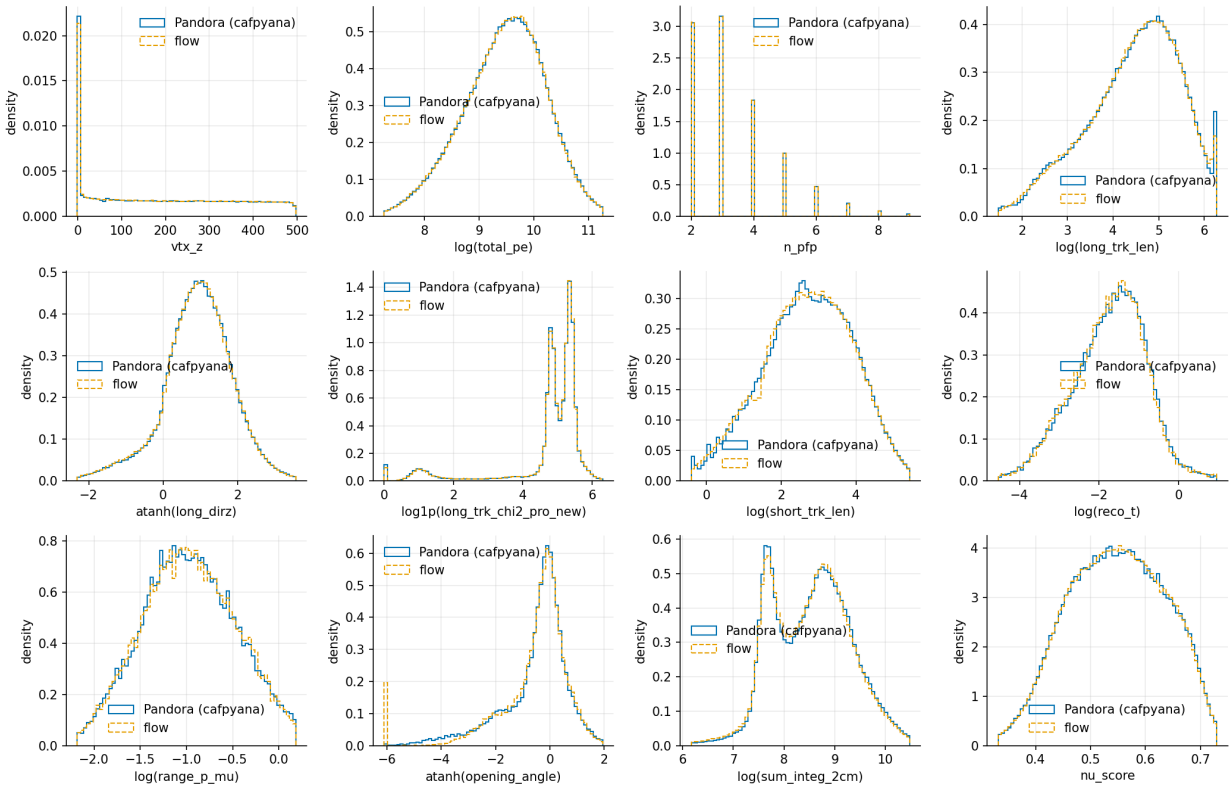


Figure 7: Marginals of generated (flow) against Pandora slice variables, test files.

matrix of 19 slice variables on 250,000 test slices: the generated matrix differs from Pandora by 0.007 on average and by at most 0.044 (all 167 defined pairs within 0.05; blank cells are pairs that are undefined because the two-track variables only exist when `n_trk`= 2). Fig. 9 shows joint densities of reco–reco pairs, including the near-deterministic range–length relation, the two-population structure of the vertex charges and of the proton chi2, and the —t——opening-angle anticorrelation that defines the cohpi signal region. Fig. 10 shows the conditional the surrogate exists to learn, reconstructed against true quantities on the same neutrinos: the exiting-muon band below the diagonal in range momentum, the mirrored band of muon/pion mis-assignment in the direction, the vertex pile-up at $z = 0$ for out-of-volume interactions, and the two populations of vertex charge.

Stage	all, Pandora	all, generated	signal, Pandora	signal, generated
pe_bcfm	453,170	454,384	483	479
2trk	38,335	38,211	226	232
dir	2,169	2,229	186	194
chi2	1,333	1,332	177	174
sphere	770	714	163	157
cathode	638	590	135	139
sr	310	289	124	119

Table 4: cohpi cut flow recomputed on real and generated test slices.

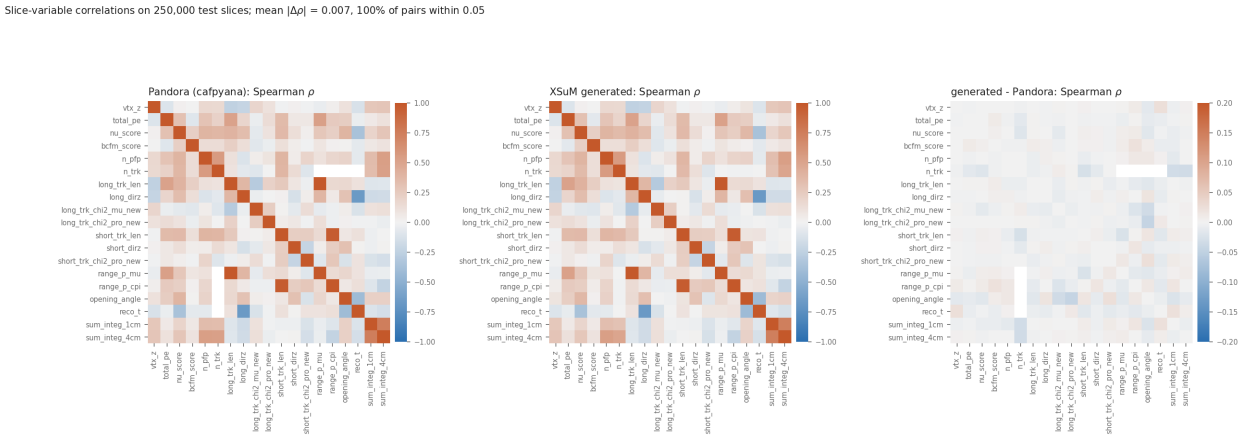


Figure 8: Spearman correlation matrices of slice variables, Pandora (left), XSuM (middle), difference (right, scale ±0.2), test slices.

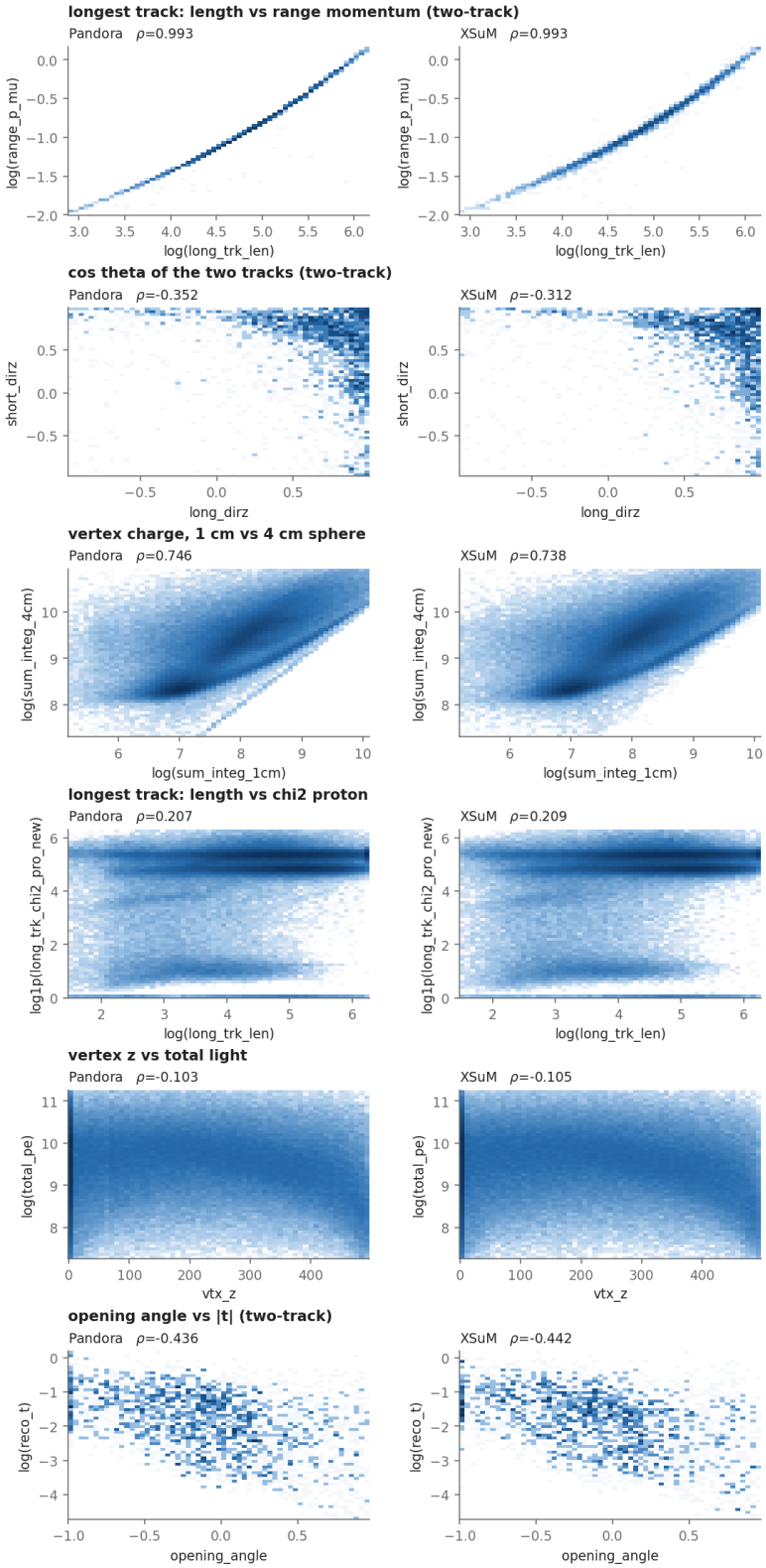


Figure 9: Joint densities of pairs of reconstructed variables, Pandora (left) and XSuM (right), same test neutrinos, same bins and color scale.

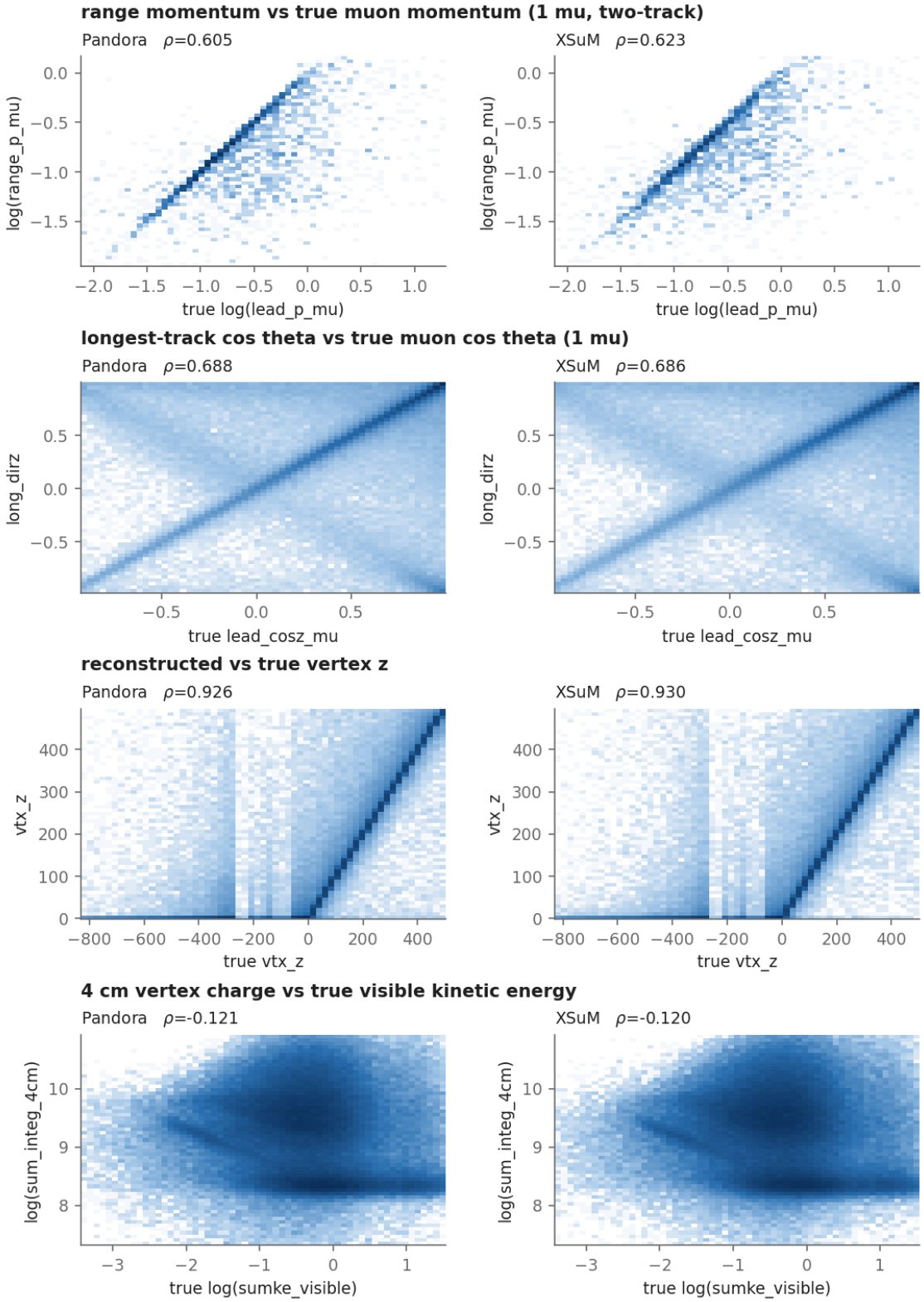


Figure 10: Reconstructed against true quantities, Pandora (left) and XSuM (right), same test neutrinos.

Column	present, real [%]	present, generated [%]	W_1 (transformed units)
n_pfp	100.0	100.0	0.0032
n_prong	100.0	100.0	0.0007
n_trk	100.0	100.0	0.0012
n_proton	100.0	100.0	0.0008
n_shower	100.0	100.0	0.0006
vtx_x	100.0	100.0	0.4322
vtx_y	100.0	100.0	0.3711
vtx_z	100.0	100.0	0.2498
nu_score	100.0	100.0	0.0005
bcfm_score	99.3	100.0	0.0017
total_pe	99.6	100.0	0.0019
long_dirx	99.9	100.0	0.0020
long_diry	99.9	100.0	0.0054
long_dirz	99.9	100.0	0.0078
long_trk_len	99.9	100.0	0.0036
long_trk_pid	99.9	100.0	0.0007
long_trk_chi2_mu	99.9	100.0	0.0100
long_trk_chi2_pro	99.9	100.0	0.0147
long_trk_chi2_mu_new	99.9	100.0	0.0099
long_trk_chi2_pro_new	99.9	100.0	0.0143
short_dirx	60.9	60.9	0.0039
short_diry	60.9	60.9	0.0067
short_dirz	60.9	60.9	0.0072
short_trk_len	60.9	60.9	0.0123
short_trk_pid	60.9	60.9	0.0026
short_trk_chi2_mu	60.9	60.9	0.0145
short_trk_chi2_pro	60.9	60.9	0.0237
short_trk_chi2_mu_new	60.9	60.9	0.0137
short_trk_chi2_pro_new	60.9	60.9	0.0225
reco_t	5.2	5.1	0.0215
opening_angle	5.2	5.1	0.0642
range_p_mu	5.2	5.1	0.0049
range_p_cpi	5.2	5.1	0.0079
muon_start_x	5.2	5.1	1.4483
muon_start_y	5.2	5.1	0.8414
muon_start_z	5.2	5.1	2.0041
muon_end_x	5.2	5.1	2.4164
muon_end_y	5.2	5.1	2.4149
muon_end_z	5.2	5.1	4.7265
cpi_start_x	5.2	5.1	1.4092
cpi_start_y	5.2	5.1	0.7554
cpi_start_z	5.2	5.1	2.0409
cpi_end_x	5.2	5.1	1.3756
cpi_end_y	5.2	5.1	0.9806
cpi_end_z	5.2	5.1	2.2161
sum_integ_1cm	86.8	86.8	0.0144
sum_integ_2cm	95.3	95.3	0.0211
sum_integ_3cm	97.2	97.2	0.0156
sum_integ_4cm	98.1	98.1	0.0203

Table 5: Every generated slice column against Pandora, test files.

Generation. `scripts/generate_slc.py` loads the trained surrogate and writes slice records for any truth table: slice cardinality (0, 1 or 2 per neutrino) from its head, then topology, presence, point masses and the flow, then the production column names, plus the recomputed cohpi flags. On the 814,028 test neutrinos of the five chunks it generates 269,841 slices in 66 s on one GPU (about 12,000 neutrinos per second, so the full official sample takes a quarter of an hour), with the slice-multiplicity fractions (0 / 1 / 2 slices: 68.0 / 31.0 / 1.1%) reproduced to 0.1% and 422 slices passing the full cohpi selection against 423 for Pandora (216 against 220 in the signal region).

6 M4: the PFP set of a slice

The particle-flow objects of a slice are generated as a set. Per matched slice the fitted PFPs (those with a track or shower fit; the few without either are counted separately) are sorted by distance to the vertex and padded to 12 (99.99% of slices have at most 12). Each PFP carries 38 fields (record layout v4): presence flags (track fit, shower fit, MCS fit, primary), distance to the vertex, track score, start point, direction, range and MCS muon momenta, chi2 for muon, proton (both calibrations), kaon and pion on the collection plane, PIDA and the chi2 degrees of freedom, the shower energy, length, opening angle, density, conversion gap and dE/dx, and the track end point: the displacement end−start (3 numbers, generated independently of the direction, so that multiple scattering can bend the chord away from the initial direction), the excess of the track length over the chord, and per axis a flag that the end lies on a TPC wall plane with the outward offset from it. On generation flagged ends are snapped onto the wall plane plus the offset (Sec. 8). Flags are dequantised, fields that do not exist for a PFP are filled with noise and restored from the flags after generation (sim2reco's prong convention). The conditioning vector (309 numbers) is the truth features with the M3 embedding, the topology one-hot and the generated slice block, so the PFP set is consistent with the slice record. Only generator truth enters: no Geant4 quantity is an input (Sec. 8). Two softmax heads give the number of fitted and of empty PFPs; a masked set flow (transformer velocity net, $d = 256$, 4 layers, over the PFP tokens, flow matching) generates the records. Trained on 3,000,000 slices for 40 epochs (45 min on the RTX 3090); this is the shipped configuration M3w0 (2026-11-08). Earlier versions (v2: 29 fields, track end derived as start + length × direction) are in the git history of this note.

The number of fitted PFPs is predicted with 93.2% accuracy (log loss 0.173 against 1.552 marginal): the slice block already carries the multiplicities. Table 7 compares every generated field with Pandora on the test slices (median W_1 0.006, maximum 0.711, in transformed units), Fig. 11 shows marginals. The physics closure is Table 6: the cohpi multiplicities (prongs, track candidates, proton candidates, showers) recomputed from the generated PFPs against the same recomputation on the real PFPs; the exactly-two-tracks topology rate is 8.07% generated against 7.97% real (7.97% from the slice table itself).

Count	mean, Pandora	mean, generated	0/1/2/3/4, Pandora	0/1/2/3/4, generated
n_prong	1.587	1.596	4344 / 161854 / 97884 / 28326 / 6499	4199 / 161344 / 97982 / 28245
n_trk	1.439	1.444	18228 / 165064 / 90105 / 22177 / 4166	18529 / 164416 / 89836 / 22416
n_proton	0.409	0.408	194597 / 90844 / 13320 / 1559 / 159	195310 / 90076 / 13168 / 1729
n_shower	0.374	0.383	222945 / 51455 / 19369 / 5085 / 1262	223121 / 49851 / 19657 / 5973

Table 6: cohpi multiplicities recomputed from PFP sets, test slices.

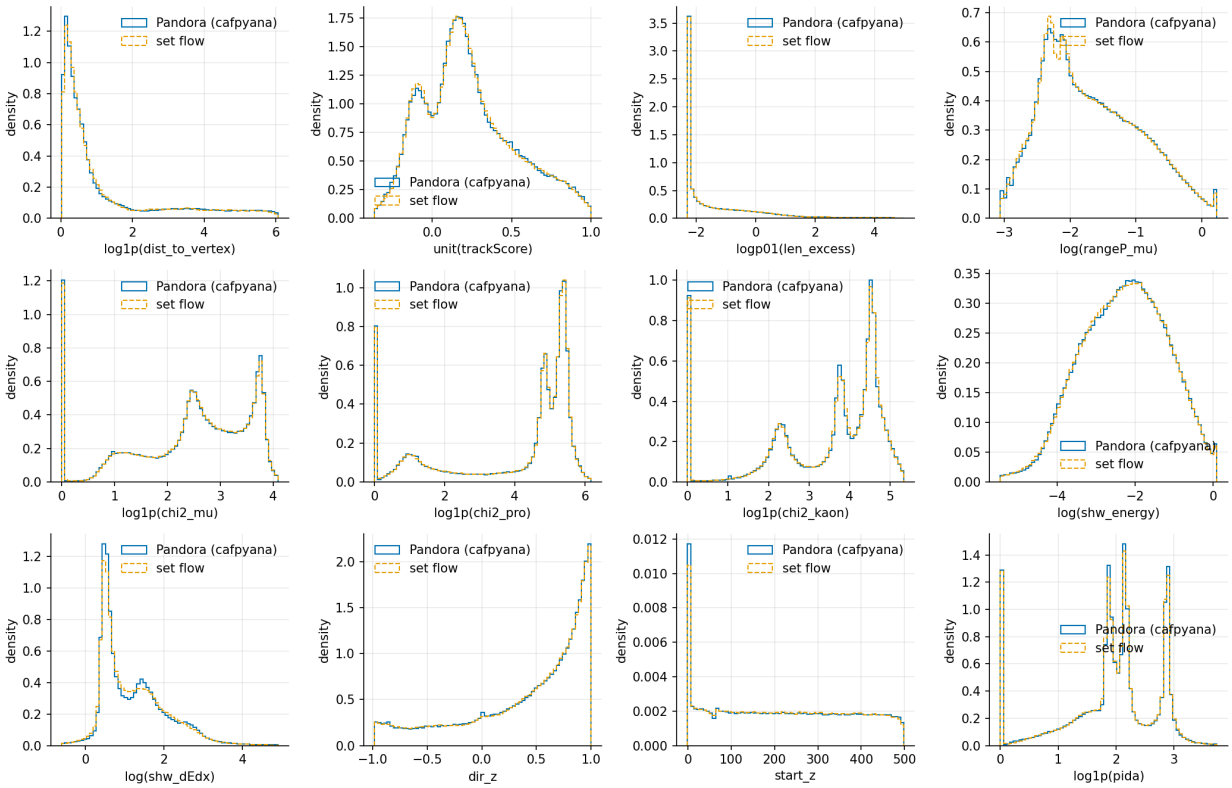


Figure 11: PFP fields, generated set against Pandora, test slices (valid tokens).

Field	Pandora median (flags: mean)	generated	W_1 (transformed units)
has_trk	1.000	1.000	–
has_shw	0.999	0.999	–
has_mcs	0.707	0.704	–
is_primary	0.927	0.931	–
dist_to_vertex	0.549	0.552	0.0175
trackScore	0.191	0.191	0.0019
start_x	-18.933	-18.983	0.5232
start_y	1.692	1.899	0.5254
start_z	221.122	221.744	0.7106
dir_x	0.008	0.010	0.0021
dir_y	0.005	0.005	0.0017
dir_z	0.604	0.605	0.0024
len_excess	-1.908	-1.902	0.0066
rangeP_mu	-1.792	-1.796	0.0023
mcsP_mu	-0.916	-0.932	0.0165
chi2_mu	2.628	2.628	0.0057
chi2_pro	4.894	4.897	0.0035
chi2_mu_new	2.629	2.627	0.0062
chi2_pro_new	4.906	4.905	0.0045
chi2_kaon	3.924	3.930	0.0056
chi2_pion	2.954	2.952	0.0060
pida	2.032	2.034	0.0065
chi2_ndof	3.332	3.299	0.0234

Field	Pandora median (flags: mean)	generated	W_1 (transformed units)
shw_energy	-2.234	-2.240	0.0039
shw_len	3.128	3.130	0.0050
shw_open_angle	0.026	0.027	0.0036
shw_density	1.593	1.597	0.0037
shw_conversion_gap	0.539	0.544	0.0170
shw_dEdx	0.973	0.981	0.0147
disp_x	0.005	0.010	0.0026
disp_y	0.006	0.003	0.0048
disp_z	0.844	0.850	0.0109
wall_x	0.055	0.052	–
wall_y	0.062	0.059	–
wall_z	0.098	0.102	–
wall_off_x	-0.077	-0.075	0.0107
wall_off_y	0.103	0.221	0.1417
wall_off_z	0.225	0.219	0.0144

Table 7: Every generated PFP field against Pandora, test slices.

7 M3: the full final-state particle list as input

Motivation. A collaborator's CC1p0pi selection (M. Fernandez), run on the test files of `surrogate_official_v3`, agreed with Pandora up to the two-prong stage and diverged at "2prong-contained" (efficiency 0.626 real against 0.387 surrogate, Table 11). That cut requires the vertex and both track end points inside the fiducial volume and on the same side of the cathode, and the signal definition requires the true muon and proton to stop in one TPC. Per-event comparison of Pandora and the surrogate on the same neutrinos located the cause: the transverse track directions were uncorrelated with the event (Pearson +0.06 for x, +0.05 for y, Table 9). The summary features of Sec. 3 hold each leading particle's $\cos\theta$ and p_T but no azimuth, so the model could reproduce the containment *rate* (Sec. 6) but not *which* events are contained, which is what a truth-defined containment signal tests.

Inputs. For every neutrino the post-FSI primaries (`mcprim`; neutrinos, nuclear fragments and the Bertini binding-energy pseudo-particle dropped) form a set of up to 32 tokens sorted by kinetic energy (0.45% of neutrinos have more and lose their softest particles). Each token has a species class (12 classes as in Sec. 3) and (p_x, p_y, p_z, T); it is embedded from the class and from its unit direction, $\log|p|$, $\log T$ and three geometric path lengths from the true vertex along its direction (to the cathode if it heads towards it, to the nearest x/y wall, to the nearest z wall). A context token carries the 80 summary features.

Encoder (stage A). A 4-layer transformer ($d = 128$, 4 heads) over the context and particle tokens, pooled to a 64-dimensional embedding, is trained on 3 M matched slices of the training files for 6 epochs on the targets that matter for selection: the topology class (cross entropy), the group presence and containment flags (binary cross entropy) and the continuous slice block (masked mean squared error); 7 min on the RTX 3090. The embedding is then frozen and appended to the summary features (144 inputs), and M2 and M4 are retrained unchanged on the new inputs (M2: 3 M slices, 60 epochs; M4: 3 M slices, 40 epochs). The dataset is then regenerated for all 17.9 M neutrinos (5.92 M slices, 14.06 M PFPs).

head	AUC (v3)	AUC (M3)	log loss (v3)	log loss (M3)
second track	0.995	0.996	0.086	0.082
two-track block	0.999	0.999	0.017	0.016
hits within 1 cm	0.776	0.808	0.330	0.314
hits within 2 cm	0.832	0.860	0.149	0.144
muon-candidate contained	0.892	0.965	0.405	0.235
pion-candidate contained	0.908	0.937	0.288	0.238
back-to-back pair	0.872	0.885	0.087	0.082
topology (121 classes)	accuracy 0.439 / 0.453		2.224	2.138
matched slices per neutrino (0/1/2+)			0.213	0.160

Table 8: M2 heads on the test files with the summary features only (v3) and with the M3 embedding added.

Effect on M2 and M4. The containment flags gain most (Table 8: muon candidate AUC 0.892→0.965), as expected; the vertex-activity groups, the topology and the number of matched slices also improve. The M2 flow validation loss drops from 0.776 to 0.677. The marginal closure of M2 and of M4 is unchanged: the median per-column W_1 of M2 is 0.0160 against 0.0144, the median per-field W_1 of M4 0.0048 against 0.0046 (the tables of Secs. 5 and 6 show the previous versions). Per event, the generated track directions and end points now follow the true event (Table 9). The cohpi signal efficiency is unchanged (Table 10).

column	v3	M3
`vtx_x`	+0.91	+0.97
`long_dirx`	+0.06	+0.57
`long_diry`	+0.05	+0.49
`long_dirz`	+0.34	+0.36
`short_dirx`	+0.01	+0.54
`short_diry`	+0.01	+0.46
`short_dirz`	+0.35	+0.38
`muon_end_x`	+0.63	+0.97
`muon_end_y`	+0.58	+0.95
`muon_end_z`	+0.91	+0.96
`cpi_end_x`	+0.95	+0.98
same sign of `long_dirx`	0.53	0.77

Table 9: Pearson correlation, on the same test slices, between the Pandora value and the generated value.

Acceptance test. The collaborator's CC1p0pi cut flow (`scripts/check_cc1p0pi_stages.py`, a transcription of their notebook) is run on the test files of grid jobs 0–199 of the real production and of both surrogated datasets. For each stage the table lists the signal efficiency, the purity and the fraction of the Pandora-selected neutrinos that the surrogate also selects (overlap; it measures per-event agreement, not just rates).

The particle-set inputs close most of the gap. At "2prong-contained" the signal efficiency rises from 0.387 to 0.543 (Pandora 0.626) and the purity from 0.243 to 0.339 (Pandora 0.431); the fraction of Pandora-selected neutrinos that the surrogate also selects rises from 0.31 to 0.42, and the gain carries through to the end of the selection (final efficiency 0.305 against 0.346, purity 0.780 against

	test files			all files		
stage	Pandora	v3	M3	Pandora	v3	M3
PE, BCFM	429	434	444	4416	4524	4560
2 tracks	206	212	209	2061	2082	2114
directions	173	175	177	1722	1697	1749
chi2	164	168	155	1611	1558	1576
sphere	149	158	140	1473	1436	1434
cathode	136	134	121	1266	1247	1245
signal region	121	119	111	1174	1099	1109

Table 10: cohpi cut flow for true cohpi-signal neutrinos (matched slices), Pandora against the two surrogate versions.

	Pandora			XSuM v3 (summary features)			XSuM M3 (particle set)		
stage	sel.	eff.	pur.	eff.	pur.	overlap	eff.	pur.	overlap
allreco	57559	0.968	0.041	0.966	0.041	0.82	0.974	0.041	0.88
vertex_in_fv	34578	0.963	0.067	0.950	0.066	0.84	0.963	0.067	0.86
nu_score	30496	0.872	0.069	0.882	0.069	0.82	0.877	0.068	0.83
2prong	11624	0.637	0.132	0.652	0.133	0.51	0.613	0.125	0.53
2prong-contained	3500	0.626	0.431	0.387	0.243	0.31	0.543	0.339	0.42
2prong-trackscore	2636	0.589	0.538	0.362	0.299	0.31	0.517	0.430	0.45
2prong-vtxdist	1938	0.465	0.578	0.292	0.333	0.26	0.411	0.476	0.40
2prong-muX	1146	0.387	0.813	0.218	0.455	0.23	0.334	0.695	0.40
2prong-mup-PID	936	0.346	0.890	0.202	0.502	0.23	0.305	0.780	0.41

Table 11: CC1p0pi cut flow on the test files of jobs 0–199 (test neutrinos: 179331, true CC1p0pi signal 2408).

0.890, previously 0.202 and 0.502). The selection is not yet closed: the surrogate still contains about 13% too few signal events and about 20% too much background at the containment cut. Sec. 8 follows this up: the track end points, the signal definition and the origin of the remaining purity gap.

8 Track end points, the CC1p0pi signal definition and the purity gap

All results in this section use the CC1p0pi cut flow on the test files of grid jobs 0–199, as in Sec. 7, and the per-neutrino containment diagnostic (`scripts/diag_cc1p_containment.py`: the two longest good tracks of the Pandora and the surrogate slice, each containment condition separately, and a generator-truth category). Model names: M3 is the M4 v2 model behind `surrogate_official_m3_pfp`; the other names are variants of M4 only, with M2 and the M3 encoder unchanged.

Resolution, not bias. On the same neutrinos, the surrogate's end point of the longest track differs from Pandora by about 20 cm (3D median) in M3, with signed medians below 0.5 cm on every axis: a resolution. It still moves events one way through the containment cut, because truth-contained signal sits inside the boundary while exiting background piles up at the TPC wall, 10 cm outside the FV. Real end points smeared with the XSuM residuals (random signs) reproduce the XSuM containment rates.

Intrinsic floor. How well can the end point be predicted from generator truth alone? Using the Geant4 truth only as a reference (`scripts/diag_end_floor.py`), for contained muons below 1 GeV the best straight-line prediction from the vertex, direction and range misses the Geant4 end by 9.9 cm (median; 1.6 cm from straggling and interactions in the length, 9.1 cm from multiple scattering in the direction). Two independent draws of a perfect surrogate differ by 14.5 cm (median, 90% quantile 40 cm). Per-event agreement below this is not possible without Geant4 information; the target is the distribution, not the event.

Signal definition (decided 2026-11-07). The notebook's signal requires the true muon and proton end points inside one TPC's FV. That boundary is sharp and set by stochastic end points: any surrogate with the floor above moves signal out and exiting background in, while the reconstructed containment cut still selects the same mixture. The primary definition is now: ν_μ CC with one muon, one proton and no pions, p_μ and p_p below 1 GeV, true vertex in the FV outside the cathode band, no end-point containment (7,686 signal neutrinos); the notebook definition (2,408) is kept as the second table.

End-point models tried. M3b (a fix of the end-point bend) and M3e (the end point as a direct target plus PFP-to-particle cross-attention) left the end-point error and the containment unchanged. M3g put the Geant4 start, end, length and containment flags on the particle tokens (error 20→7 cm) and M3w added the Geant4 straight-line TPC exit point and the wall flag. Both are *not admissible*: a surrogate for new generator models cannot use Geant4 quantities as inputs (decision 2026-11-07); they are kept only as diagnostics. The admissible pieces are the decoupled displacement and the wall flag and offset of layout v4 (Sec. 6). These are reconstruction-level targets: 21% of real tracks end on a wall, sharply at $x = \pm 201.3$ cm and $z = 0$, 500.5 cm, in a band at $|y| \approx 199$–204 cm.

Particle cross-attention, ablation. 1 M slices, 15 epochs, no Geant4 inputs, the same test slices (Table 12). Two layers gain 1 cm in the per-event end-point distance and reach the two-draw floor; four layers add nothing; split by generator-truth class (CC Nπ, CC 0π 1p, CC 0π other, NC), the end-in-FV and on-wall rates do not change at all. Attention doubles the parameters and the training time and slows generation, so the shipped M4 has none. The straight line from the generated start, length and direction is closer to Pandora per event (12.8 against 14.3 cm) because it has no lateral scatter (real 7.9 cm), but it cannot end on the wall; the decoupled end point reproduces both.

particle attention layers	0	2	4	Pandora
longest-track end, |gen−real| median [cm]	15.3	14.3	14.2	14.5 (floor)
straight line start + len·dir [cm]	13.0	12.8	12.8	
lateral deviation from the straight line [cm]	9.0	8.8	8.6	7.9
end in the FV	0.470	0.469	0.472	0.497
end on the wall (±3 cm)	0.355	0.363	0.359	0.349
validation loss	0.750	0.734	0.733	
parameters	2.2 M	4.4 M	5.5 M	
training time	7 min	20 min	25 min	

Table 12: PFP-to-particle cross-attention ablation (no Geant4 inputs), longest track per test slice.

The shipped M4 (M3w0). No attention, layout v4, generator truth only, 3 M slices, 40 epochs. Longest-track end against Pandora: 11.6 cm per event (straight line 10.0), lateral deviation 7.8 cm (Pandora 7.9), end in FV 0.486 (0.497), on the wall 0.363 (0.349), outside the TPC 2.2% (0.2%). Per background category the wall pile-up is now reproduced (CC Nπ longest track on the wall 0.36 against 0.43 real; M3 0.09, with 17% of its ends outside the TPC), and when the real longest track ends on the wall the generated one does too in 90% of events (M3w with Geant4 inputs: 95%). Table 13 gives the acceptance.

	selected	primary eff / pur	notebook eff / pur	final primary eff / pur
Pandora	3,500	0.198 / 0.435	0.626 / 0.431	0.109 / 0.895
M3	3,852	0.199 / 0.397	0.543 / 0.339	0.106 / 0.861
M3w (Geant4 inputs, diagnostic)	3,698	0.190 / 0.395	0.569 / 0.370	0.104 / 0.876
M3w0 (shipped)	3,725	0.191 / 0.394	0.531 / 0.343	0.106 / 0.872
hybrid: Pandora slices + M3w0 PFPs	3,500	0.194 / 0.426	0.559 / 0.384	0.112 / 0.880
M2w: M2 with wall targets + M3w0	3,639	0.185 / 0.390	0.515 / 0.341	0.106 / 0.870

Table 13: CC1p0pi at "2prong-contained" (selected, efficiency and purity under both signal definitions) and at the end of the selection (2prong-mup-PID), test files of jobs 0–199.

Where the purity gap comes from. Every M4 variant sits at purity 0.39–0.40 against 0.435, whatever its wall handling. The excess is background passing containment: CC Nπ (contained fraction of its 2prong events 0.347 against 0.273), NC (0.662 against 0.643), other 0π (0.379 against 0.358). Three checks locate it:

- In background events that both chains call 2prong, the on-wall rate of the longest track now agrees (0.53 against 0.54) yet more pass containment (0.155 against 0.128): it is not the wall in M4.
- The CC1p0pi 2prong and containment cuts applied inside the M4 closure (real slice block as condition, per generator-truth class) give slightly *fewer* contained background events than Pandora (CC Nπ 0.282 against 0.300, NC 0.630 against 0.675): M4 alone does not produce the excess.
- A hybrid dataset, Pandora slice table with M3w0 PFPs, recovers the purity (0.426 against 0.435; CC Nπ 0.286, NC 0.626): about three quarters of the gap comes through the M2 slice block that M4 is conditioned on.

In the M2 slice block the excess is not in one variable: binned in any slice variable, the containment probability is higher in every bin, and the bin populations explain at most 100 of the 370 extra contained background events. The slice block misses the same geometry M4 missed (Table 14): an exiting longest track has a length equal to the path from the vertex to the wall along its direction, and an exiting cohpi muon ends on the wall. M2's flow smears both, and M4 then builds tracks of the generated length that stop inside.

M2 with wall targets (M2w). The M4 treatment was applied to M2. A flag and an offset are added for the longest-track length against the path to the wall (flag when $|L/\ell_{\text{wall}} - 1| < 0.03$). The cohpi muon and pion-candidate end points get a per-axis flag and an outward offset on the wall planes. The flags are dequantised in the flow and snapped after generation; M4 is unchanged (M3w0). The slice-block wall marginals now agree with Pandora (Table 14), but the acceptance

does not move (Table 13): purity 0.390 against 0.394 for M3w0. The excess of contained background drops only from 370 to 324 events, and the final-stage numbers are unchanged. M2w also puts 4.2% of cohpi muon ends outside the TPC (Pandora 0.06%), because unflagged ends are not clipped.

Where the excess really is. With the primary signal definition (no true containment), the containment of the longest track closes for every truth category taken as a whole (CC Nπ end in FV 0.514 against 0.513, NC 0.810 against 0.815, 1μ1p 0.403 against 0.403). It fails only in the 2prong subset of CC Nπ (0.518 against 0.440; hybrid 0.454). 2prong is set by the slice PFP count (93% of $n_{\mathrm{pfp}} = 3$ slices are 2prong) and M4 follows it. In Pandora CC Nπ, slices with few PFPs have more exiting longest tracks (end in FV 0.439 at $n_{\mathrm{pfp}} = 3$, 0.554 at 5); M2 generates the exit flat in n_{pfp} (0.49–0.51). For 1μ1p the 2prong topology is 73% of the events, so no tail is selected and the same defect does not show. The NC excess is a 5% surplus of 2prong events, with containment closing.

M2t: topology sampled after the flow. A study in which the topology class is drawn after the flow from the truth features and the generated slice block (presence and spike heads on truth only, flow not conditioned on the topology), with the range momenta flagged on the length–momentum curve. Purity rises to 0.404 (efficiency 0.192), but through fewer background 2prong events (CC other 0π 2,088 against 2,278), not through the targeted link: CC Nπ at $n_{\mathrm{pfp}} = 3$ stays at 0.494. On the M2 test slices of truth CC events with a charged pion (24.6k, vertex in FV) the topology head reproduces the Pandora link when fed the Pandora slice block (P($n_{\mathrm{pfp}} = 3$ — end in FV / outside) 0.107 / 0.135 against Pandora 0.108 / 0.135) and loses it on the generated block (0.108 / 0.112). Replacing one group of the Pandora block by the generated values (`scripts/diag_m2_swap.py`) reduces the link most for the short-track group (from +0.028 to +0.016, mostly through whether a short track exists), less for the two-track group, the direction and the length; no single group restores it. The link is carried jointly by the longest track, the presence of the optional groups and the PFP counts, which M2 samples in separate pieces.

M2s: two-stage M2. Stage 1 is one flow over the always-present variables, the five PFP counts (dequantised integers) and the group-presence flags; stage 2 is a flow over the optional groups, conditioned on the truth and the stage-1 output and masked by the generated presence. The hard relations of the Pandora reconstruction (counts nested, a short track whenever $n_{\mathrm{trk}} \geq 2$, the two-track block only for $n_{\mathrm{prong}} = n_{\mathrm{trk}} = 2$) are enforced after stage 1; they correct 0.03–0.9% of the slices. The targeted link closes: CC Nπ at $n_{\mathrm{pfp}} = 3$, longest track ending in the FV 0.442 against Pandora 0.439, and the pion range momentum lies on the length curve for 93% of the slices (Pandora 93%). The chain loses 2prong events, however: efficiency 0.165 and purity 0.396 at 2prong-contained (Pandora 0.198 / 0.435), P(PFP 2prong — slice $n_{\mathrm{pfp}} = 3$) 0.85 against 0.97 for signal. The dequantised counts do not keep the joint count lattice: 17% of the generated slices carry a $(n_{\mathrm{pfp}}, n_{\mathrm{prong}}, n_{\mathrm{trk}}, n_{\mathrm{proton}}, n_{\mathrm{shower}})$ tuple that Pandora slices do not populate, and the (3, 2, 2, ·, 0) tuples are 0.341 of $n_{\mathrm{pfp}} = 3$ against 0.439, so M4 builds fewer two-track sets. A variant with the counts entering stage 1 as the topology class (a dequantised one-hot, decoded by argmax; `--topo-onehot`, pipeline M2o) is being trained. None of these studies is shipped; the shipped M2 is unchanged.

all matched slices, jobs 0–199	Pandora	M2	M2w
longest track: length / path to the wall within ±2%	0.145	0.060	0.163
length / path > 1.05	0.076	0.133	0.091
cohpi muon end on the wall (±3 cm)	0.429	0.079	0.434
cohpi muon end outside the TPC (< −3 cm)	0.001	0.187	0.043

Table 14: Wall consistency of the slice block (`scripts/diag_m2_purity.py`).

9 Surrogated datasets for the analysers

`scripts/write_surrogate_df.py` turns the generated slices into files in the production format, so that existing cafpyana-based analysis code runs on them unchanged. The first dataset, `surrogate_v3_6ch`, applies the six-chunk M2 v3 model to the 9,774,979 true neutrinos of the scratch production (the training sample itself; comparisons with the real chain must use the test files listed in `file_splits.parquet`, 10% of the input files chosen by file):

- location: `/pnfs/sbnd/persistent/users/lwan/dataframe/surrogate_v3_6ch/`, six files `xsum_surrogate_c` (11 GB), world-readable; the real production it mirrors is in `prod_v10_14_02_03_scratch1633/` (18 GB) next to it;
- keys per file: `mcnu_0, mcprim_0, hdr_0, histpotdf_0, histgenevtdf_0, pot_0` copied from the production, `slc_0` generated (index `(__ntuple, entry, rec.slc..index)`, `tmatch_idx` = generating neutrino, `tmatch_eff` / `tmatch_purity` NaN, the cohpi `pass_*` and `in_sr` flags recomputed from the generated variables, plus `surrogate_topology_class`), `provenance`;
- content: 3,241,308 generated slices (13 min of generation), of which 5,252 pass the full cohpi selection and 2,759 are in the signal region; no `pfp` table yet (M4), no cosmic-only slices (to be taken from the MC).

The second dataset, `surrogate_official_v3` (2026-10-22), applies the official-sample model of Sec. 5 to all 17,861,754 neutrinos of the official production: 5.9 M generated slices in 27 min, 6,302 of which pass the full cohpi selection against 6,095 in the real chain (2,988 in the signal region). It is written as 2,000 files `xsum_surrogate_N.df` (75 GB) in `/pnfs/sbnd/persistent/users/lwan/dataframe/surroga` one per grid job of the archive `prod_v10_14_02_03_official/`, with `mcnu, mcprim, mcdaught, pfp, hdr` and the POT tables copied from the production and `slc_0` generated (same conventions as above); `file_splits.parquet` marks the 74,533 test files. Note that the real `slc` tables of that archive must be read from the repaired sidecars `prod_v10_14_02_03_official_slcfix/` (Sec. 2).

`surrogate_official_v3_pfp` (2026-10-22) is the same dataset with a generated `pfp_0` table from the M4 v2 set model of Sec. 6: 14,005,625 PFPs for the 5,926,843 generated slices (69 min of generation), with the 26 generated fields laid out under the production column names, straight-line track end points, no PFPs without a fit, and no induction-plane chi2 or truth-match columns. This is the first dataset in which every reconstructed object of the slice is surrogated.

`surrogate_official_m3_pfp` (2026-11-05) is the same construction with the M2 and M4 models retrained on the particle-set inputs of Sec. 7: 5,923,999 generated slices and 14,061,560 PFPs, 2,000 files in `/pnfs/sbnd/persistent/users/lwan/dataframe/surrogate_official_m3_pfp/`. It supersedes `surrogate_official_v3_pfp` for any selection that uses track directions or containment.

`surrogate_official_m3w0_pfp` (2026-11-08) replaces the PFP table with the shipped M4 (M3w0, Sec. 8): generated track end points with the wall pile-up, generator truth only; same M2 slices (5,923,999) and 14,061,560 PFPs (94 min of generation). Written so far for the test files of jobs

0–199 in `/pnfs/sbnd/persistent/users/lwan/dataframe/surrogate_official_m3w0_pfp/`; the other 1,800 jobs follow after review.

10 Outlook

Done so far: the official sample, M1 baselines, M2 (slice block), M3 (particle-set encoder, stage A), M4 (PFP set, generated track ends on the wall), and production-format surrogated datasets for the analysers. Remaining:

1. Close the CC1p0pi purity gap (Sec. 8): the two-stage M2 (M2s) closes the CC Nπ n_{pfp}–exit link but breaks the joint PFP counts; the version with the counts as a one-hot topology class in stage 1 (M2o) is in training.
2. Write `surrogate_official_m3w0_pfp` for all 2,000 jobs once reviewed.
3. Range momenta from the generated lengths: `range_p_mu` and `range_p_cpi` are functions of `long_trk_len` and `short_trk_len` (97% / 93% of Pandora slices within 2%), yet M1 and M2 generate them as independent numbers and smear the sharp range peak of stopping pions (Fig. 6; M2 peak fraction 0.17 against 0.22). M2t flags them on the curve and snaps them, but reaches only 72% on the curve for the pion; the two-stage M2s reaches 93% (Pandora 93%). To be shipped with the next M2, with the M2 version of that figure.
4. M4 completeness: induction-plane chi2, PFPs without a fit, PFP truth-match columns.
5. Cosmic-only and unmatched slices in the surrogated files (taken from the MC).
6. Uncertainty: Bayesian last layers or ensembles, an out-of-support flag per generated record.
7. Holdouts: train on $|t| < 0.1$, test the tail; low versus high E_ν; forward versus wide-angle muons; ≤ 2 versus ≥ 3 charged primaries; one TPC versus the other.
8. Collaborators' comparisons (cohpi, kaon, CC1p0pi) on the test files of `surrogate_official_m3w0_pfp`.